

EXERCISE 3:

Multimedia Databases SS 26

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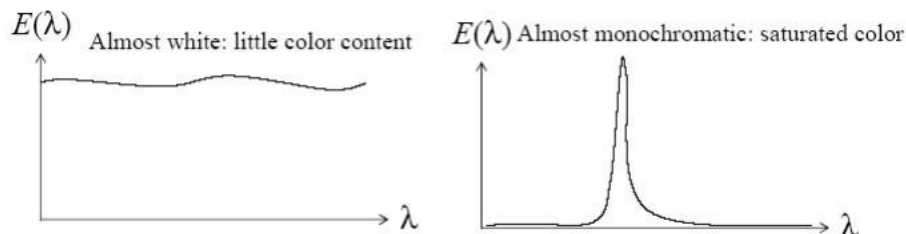
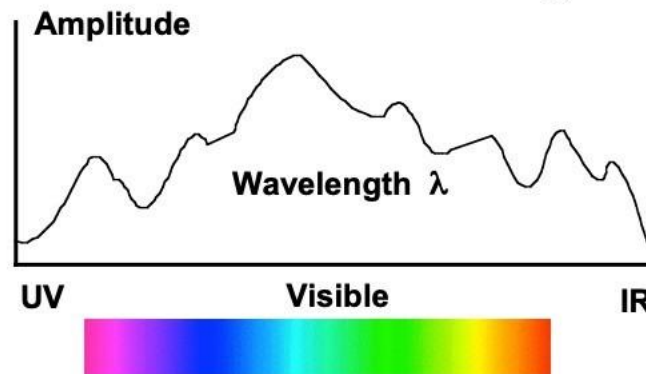
Task 1.1:

Robert Collins
CSE486, Penn State

Color of Light Source

Spectral Power Distribution:

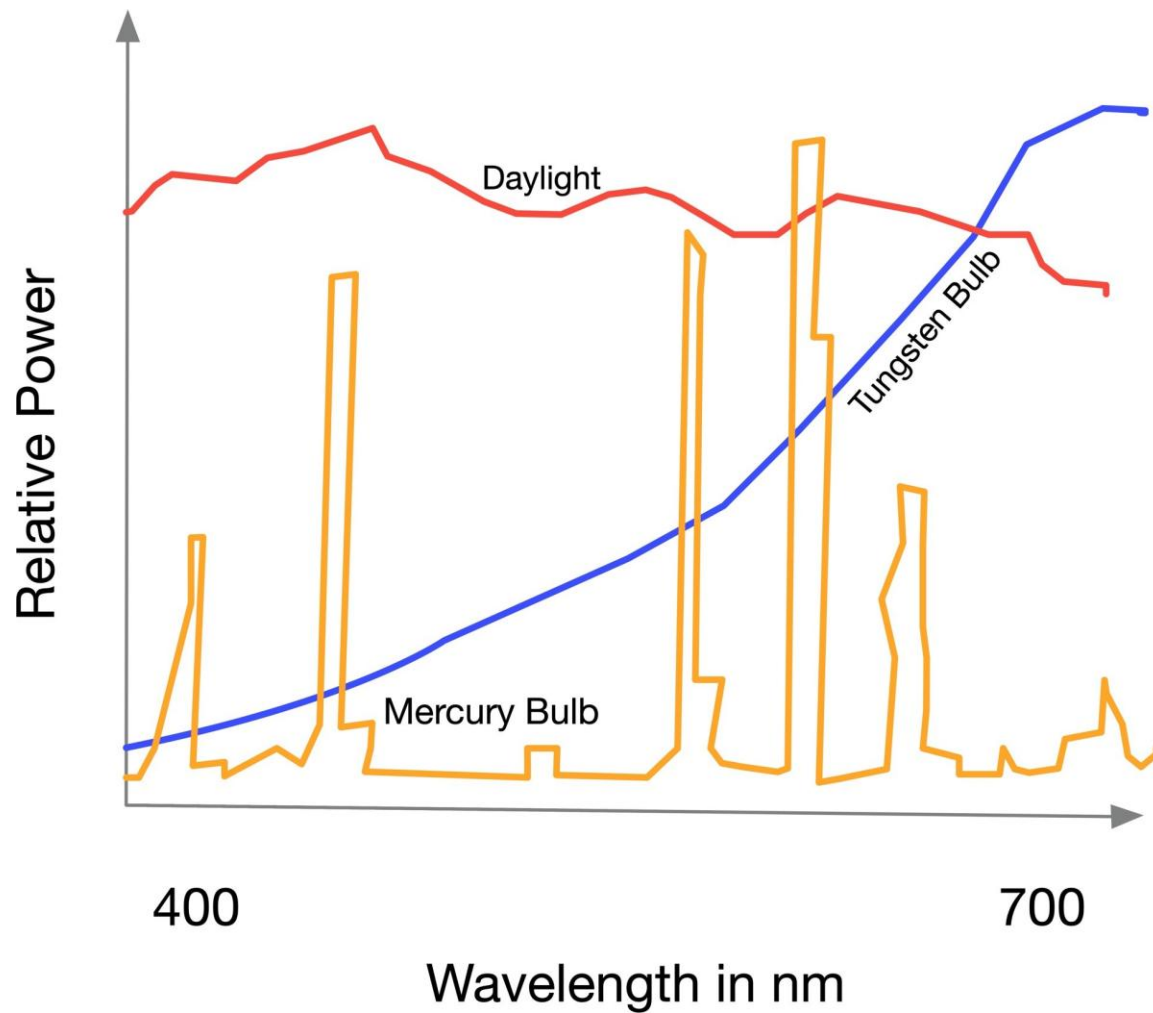
Relative amount of light energy at each wavelength



Source: <http://www.cse.psu.edu/~rtc12/CSE486/lecture26.pdf>

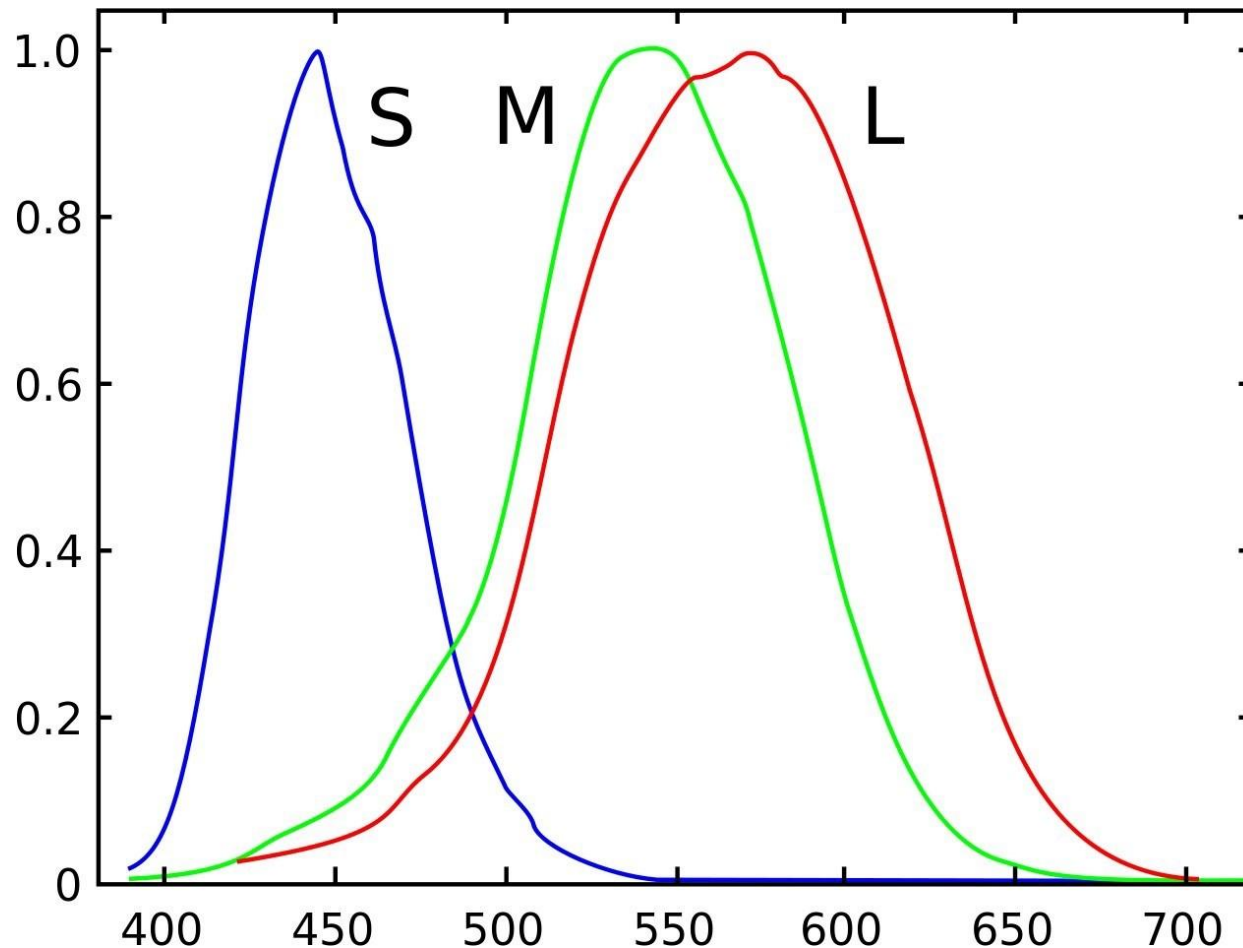


Task 1:1



Task 1.2

Spectral sensitivity is the relative efficiency of detection, of light or another signal, as a function of the frequency or wavelength of the signal.



Task 1.3: Metamers

Color stimuli that have different spectral radiant power distributions but are perceived as identical for a given observer i.e. different spectral profiles that produce exactly the same *relative* stimulation to the **L**, **M** and **S** cones

- **light metamers**, different spectral emittance curves perceived as the same color
- **material metamers**, two different surface reflectance curves perceived as the same color when each is viewed with the same light source
- **observer metamers**, different spectral profiles perceived as the same color due to limitations in the viewer's visual responses (colorblindness or dark adapted vision).



1.4: Chromatic Adaptation

- In everyday life, illumination sources are changing constantly (full sunshine, cloudy conditions, indoor lighting etc)
- Measured spectrum of light reflected from a given object is constantly changing.
- Despite this, we perceive that “the light changed” not “all the objects changed color”.
- Rather like a camera with auto white balance, our eye constantly recalibrates to what it sees as “white light”, and all other colors are judged relative to that white.



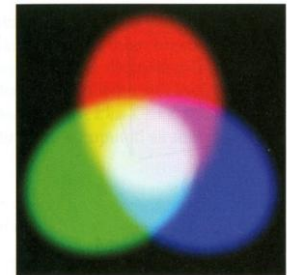
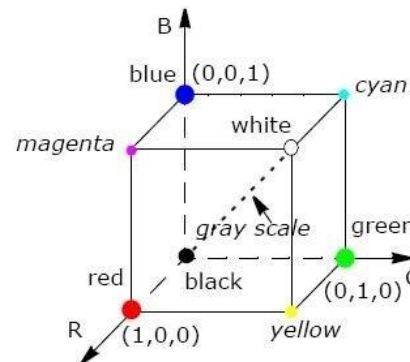
2.1: Color Model

- **What is a color model?**
 - Abstract Method for representing color information
 - Makes use of the characteristics of the human vision system.
 - Example: RGB represents a color as three dimensional vector
- **Two types: Additive & Subtractive Color model**



2.2: Additive Color Model

- **Lights are used to produce colors.**
 - Black represents the complete lack/absence of the primary colors.
 - White corresponds to **maximal** and **equal amounts of each of the primary colors**.
 - E.g.; RGB model
 - Black $\rightarrow (0,0,0)$ White $\rightarrow (1,1,1)$ s.t. **0** indicates absence of the primary color and **1** maximum amount of the specified primary color.
 - Additive primary colors:
 - Red, Blue, Green
 - Additive secondary colors:
 - Cyan, Magenta, Yellow
 - Complementary color pairs:
 - Red-cyan
 - Green-magenta
 - Blue-yellow

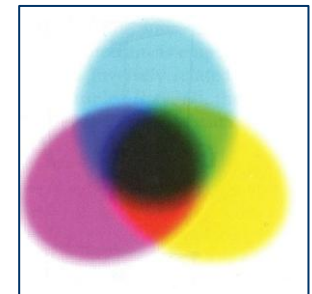
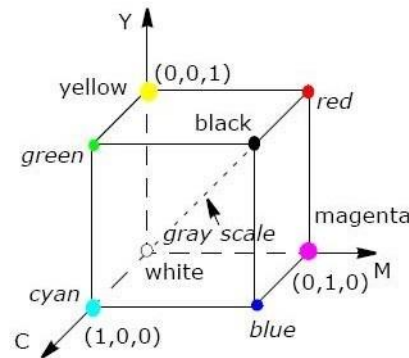


RGB model (additive)

- **Human perception is an additive system.**
- Hardwares/ Devices naturally modeled using an additive color model: television, computer monitors, LCD projectors, stage lighting, etc.

2.2: Subtractive Color Model

- **Pigments or dyes are used to produce colors.**
 - White represents the complete lack/absence of any pigment.
 - Black corresponds to **maximal** and **equal amounts of each of the primary colors**.
 - E.g.; CMY model
 - White $\rightarrow (0,0,0)$ Black $\rightarrow (1,1,1)$ s.t. **0** indicates absence of the primary color and **1** maximum amount of the specified primary color.
 - Subtractive primary colors:
 - Cyan, Magenta, Yellow
 - Subtractive secondary colors:
 - Red, Green, Blue
 - Complementary color pairs:
 - Magenta–green
 - Yellow–blue
 - Cyan–red

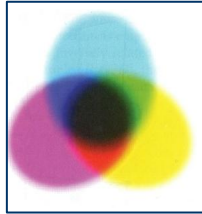


CMY model
(subtractive)

- Subtractive color models tend to be more intuitive than additive ones.
- Hardwares/ Devices naturally modeled using a subtractive color model: ink-based printers, etc.

2.2: Color model

• CMY (K) color model:



- Subtractive color model
- CMY(K)= Cyan (1,0,0) Magenta (0,1,0) Yellow (0,0,1)
- CMY model is inversely related to the RGB model

- $C = 1 - R$

- $M = 1 - G$

- $Y = 1 - B$

$$\langle C, M, Y, K \rangle = \begin{cases} \langle 0, 0, 0, 1 \rangle & \text{if } \min(C', M', Y') = 1, \\ \langle \frac{C'-K}{1-K}, \frac{M'-K}{1-K}, \frac{Y'-K}{1-K}, K \rangle & \text{otherwise where } K = \min(C', M', Y'). \end{cases}$$

- What is the k?
 - K stands for „key“
 - (1,1,1) -> "pale black,,
 - The black is used in the printer industry to achieve a deeper black than could be made by mixing the three primaries. It is also used to save printer ink.
- The conversion from CMY to CMYK:

2.2: Color model

- **HSB(V) color model:**

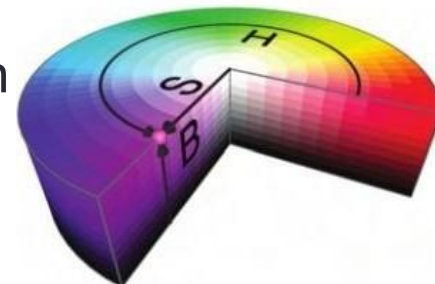
- **HSB = Hue Saturation Brightness**

- Hue = color (0-360)
- Saturation = degree to which the hue differs from a neutral gray (0 no color saturation -1 full saturation)
- Brightness = illumination level, from 0 (black, no light) to 1 (white, full illumination).

- Designed to approximate the way humans perceive and interpret color.

- **Advantages:**

- (HS) and (B) treated separately
- Complementary colors by shifting the h
- Digital image processing



2.3: Converting from RGB to CMYK

- Given a colour represented in RGB colour space as $R = 0.2$, $G = 0.6$, $B = 0.3$, what is its representation in the CMYK colour mode?

$$\begin{pmatrix} \bar{C} \\ \bar{M} \\ \bar{Y} \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - \begin{pmatrix} R \\ G \\ B \end{pmatrix} = \begin{pmatrix} 0.8 \\ 0.4 \\ 0.7 \end{pmatrix}$$

$$K = \min(\bar{C}, \bar{M}, \bar{Y}) = 0.4,$$

$$C = \frac{\bar{C} - K}{1 - K}, M = \frac{\bar{M} - K}{1 - K}, Y = \frac{\bar{Y} - K}{1 - K}$$

$$C = 0.67, M = 0, Y = 0.5$$



2.4: RGB to HSV

Color models Converting RGB to HSV

► **Conversion RGB $\rightarrow$ HSV** (from Gonzalez and Woods)

1. Compute White – ratio $W = \min(R, G, B)$
2. $(R', G', B') := (R - W, G - W, B - W)$ has only 2 color values $\neq 0$
3. Let $B' = 0$: Hue – the value lies between R and G:

$$H = G' * \frac{120}{R' + G'}$$

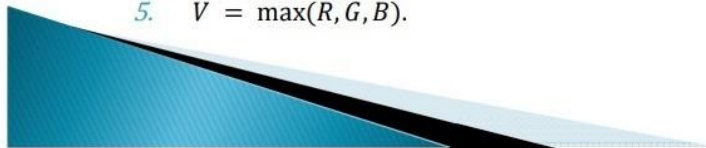
Let $R' = 0$: Hue – the value lies between G and B:

$$H = B' * \frac{120}{G' + B'} + 120$$

Let $G' = 0$: Hue – the value lies between B and R:

$$H = R' * 120 / (R' + B') + 240$$

4. $S = (\max(R, G, B) - W) / \max(R, G, B)$
5. $V = \max(R, G, B)$.



2.4: RGB to HSV

$$W = \min(R, G, B) = \min(0.2, 0.6, 0.3) = 0.2$$

Therefore,

$$(R', G', B') = (R - W, G - W, B - W) = (0, 0.4, 0.1)$$

Since $R' = 0$,

$$H = (0.1 \times 120 / 0.5) + 120 = 144 \text{ degrees}$$

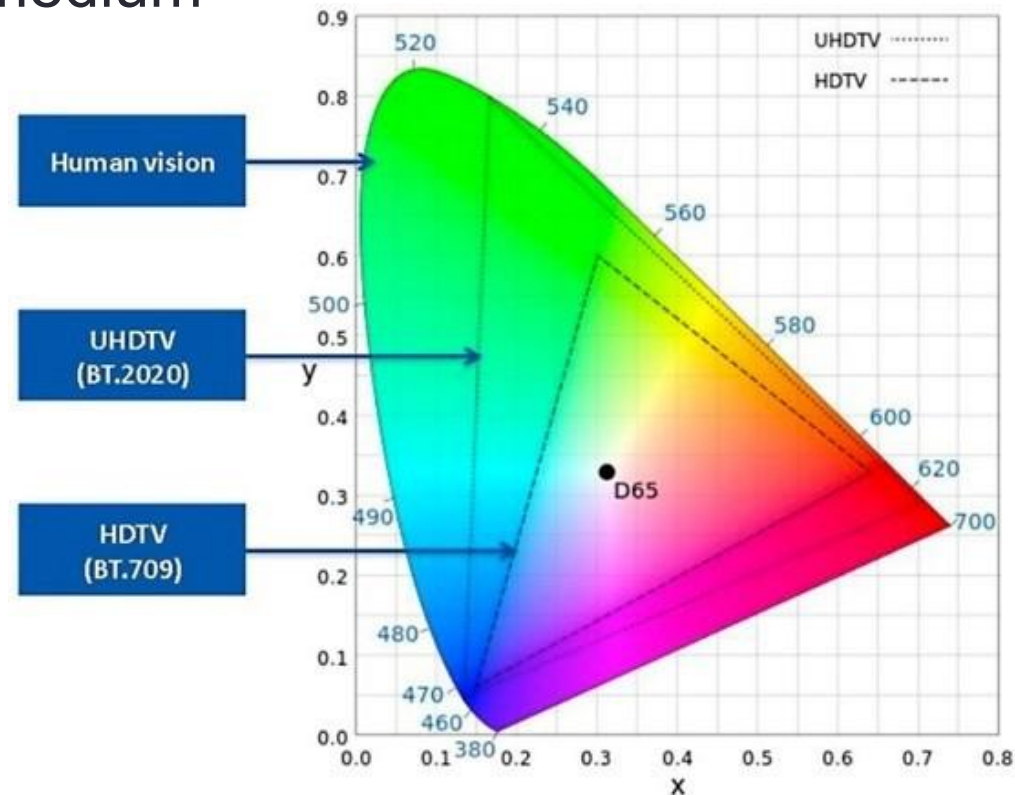
$$S = (0.6 - 0.2) / 0.6 = 0.66 = 66.7\%,$$

$$V = 0.6 = 60\%.$$



3: Color Space

- A **color model** is a method of describing a color, a **color space** is the set of colors which can be displayed or reproduced in a medium



3.1: Color Space

There are multiple ways of doing the YCbCr $\leftrightarrow$ RGB [conversion](#). The general conversion is given by

$$\begin{aligned} Y &= K_r * R + K_g * G + K_b * B \\ C_b &= (B - Y) / (1 - K_b) \\ C_r &= (R - Y) / (1 - K_r) = R - G * K_g / (1 - K_r) - B * K_b / (1 - K_r) \end{aligned}$$

and the other way around:

$$\begin{aligned} R &= Y + C_r * (1 - K_r) \\ G &= Y - C_b * (1 - K_b) * K_b / K_g - C_r * (1 - K_r) * K_r / K_g \\ B &= Y + C_b * (1 - K_b) \end{aligned}$$

where

$$\begin{aligned} (0.0 \leq [Y, R, G, B] \leq 1.0), \\ (-1.0 \leq [C_b, C_r] \leq 1.0) \text{ and} \\ K_r + K_g + K_b = 1 \end{aligned}$$

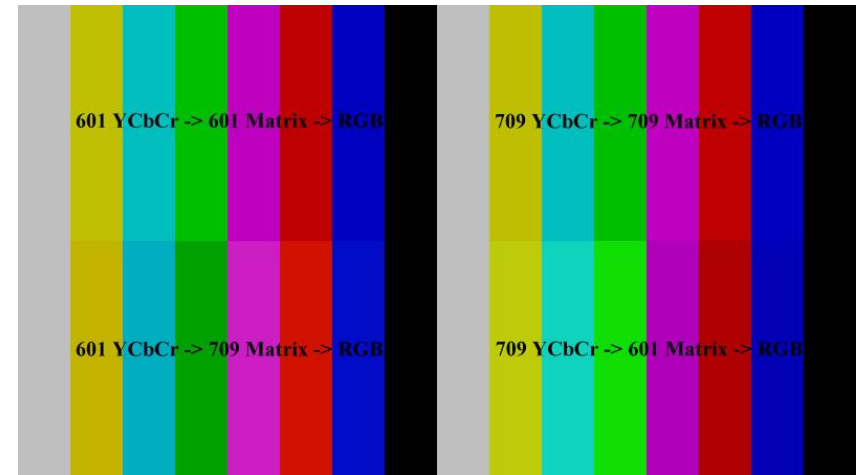
The red, green and blue coefficients (K_r, K_g, K_b) are standardized in specifications. The most common conversion coefficients are given below:

Standard	K_r	K_g	K_b
Rec.601/BT.601 ⁽²⁾  , SMPTE 170M ⁽³⁾ 	0.299	0.587	0.114
FCC ⁽²⁾ 	0.300	0.590	0.110
Rec.709/BT.709 ⁽²⁾ 	0.2126	0.7152	0.0722
SMPTE 240M ⁽⁴⁾ 	0.212	0.701	0.087



3.1: Color Space

- **Color Model of Video:**
 - YCbCr
- **Conversation on Playback:**
 - YCbCr $\rightarrow$ RGB
- **Different coefficients on conversation**
 - **BT.601** Typically on SD Video
 - **BT.709** Typically on HD Video
 - **BT2020** Typically on HDR Video
- **What went wrong with 2nd video?**
 - Video in YCbCr(BT.709)
 - Player chose coefficients based on resolution (BT.601)



3.2: CIE XYZ vs CIE L*a*b

Adapted from: <http://web.mit.edu/6.813/www/sp18/classes/15-color/>

CIE XYZ	CIE L*a*b
• CIE XYZ is an <i>additive, linear light</i> colorspace useful for calculating color mixtures, but is not perceptually uniform. • The distance apart of two colors in XYZ does not directly relate to how similar or different they appear. • <i>Chromatic adaptation</i> is also not well shown in XYZ.	• Oriented towards the physiological properties of human (color) perception. • The distance between two colors in Lab color space accurately predicts how similar or different they look. • All colors are chromatically adapted to a color temperature of 5000K (D50) so that measurements made under different conditions can be compared with each other.

